The Journal of Open Source Software

RCAEval: A Framework for Multimodal Root Cause Analysis

Luan Pham [1]

1 RMIT University, Australia

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 09 December 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

RCAEval is an open-source framework for multimodal root cause analysis (RCA). It provides 20+ ready-to-use RCA methods including single-modality and multi-modality approaches. It is accompanied by comprehensive datasets containing 735 failure cases from real-world systems. RCAEval is a Python framework, open-source on GitHub, and installable via PyPi. The datasets are available in Zenodo which can be programmatically downloaded. RCAEval provides tools and datasets, enabling researchers to evaluate RCA methods under realistic conditions.

Statement of need

Modern large and dynamic systems generate massive amounts of observability data including time series metrics, textual logs, and topological/graphical tracing. When failures occur, they can propagate across multiple components, making it challenging for operators to identify root causes from the overwhelming volume of observable data. Root cause analysis (RCA) aims to pinpoint the faulty component and the specific indicators responsible for the failure.

Despite growing research interest in automated RCA, the field lacks a standardized, reproducible benchmark. Existing studies typically evaluate on limited systems with few fault types, often using private datasets that prevent fair comparison. Available resources provide only single-modality analysis (e.g., metrics only) without support for multimodal data combining logs and traces. Commercial observability platforms offer RCA capabilities but are proprietary and not reproducible for research purposes.

RCAEval provides a unified framework with 20+ reproducible RCA method implementations including several state-of-the-art methods, e.g., BARO (Pham et al., 2024a), CausalRCA (Xin et al., 2023), a standard evaluation pipeline, and comprehensive datasets with 735 failure cases. The benchmark datasets were published in my prior works (Pham et al., 2024a, 2024b, 2025) on microservice systems. This paper focuses on describing RCAEval as a framework for the general RCA task, which is extendable and applicable to various domains. RCAEval targets researchers developing new RCA algorithms, practitioners evaluating methods for production deployment, and educators teaching algorithms.

State of the field

Several tools and datasets exist for RCA, but none provide comprehensive coverage of multimodal telemetry with reproducible methods. Existing libraries focus on metric-based RCA, supporting methods like Bayesian networks and Granger causality, but lack support for log and trace analysis. Available datasets provide limited fault types and no benchmarking framework for systematic evaluation. Commercial observability platforms offer automated root cause analysis features, but their proprietary nature prevents reproducible research comparisons.

Overview

RCAEval distinguishes itself by providing the first open-source benchmark framework that combines 20+ reproducible RCA tools with large-scale multimodal failure datasets. This enables fair, systematic comparison of RCA methods under realistic failure scenarios. The Figure 1 gives an overview on the RCAEval framework.

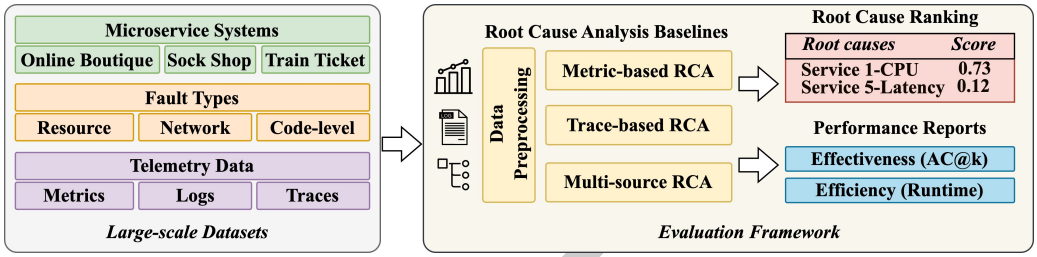

Figure 1: Overview of RCAEval benchmark framework.

Example

RCAEval provides a simple Python API for running RCA methods. The following example demonstrates using BARO, a metric-based RCA method, to identify root causes from telemetry data:

```python
from RCAEval.e2e import baro
from RCAEval.utility import download_data, read_data

# Download sample telemetry data
download_data()

# Read data and specify anomaly timestamp
data = read_data("data.csv")
anomaly_detected_timestamp = 1692569339

# Perform root cause analysis
results = baro(data, anomaly_detected_timestamp)

# Get ranked root causes
print(results["ranks"][:5])
# ['emailservice_mem', 'recommendationservice_mem',
#  'cartservice_mem', 'checkoutservice_latency', 'cartservice_latency']
```

Download Datasets

RCAEval provides utility functions to download the benchmark datasets programmatically:

```python
from RCAEval.utility import (
    download_re1_dataset,
    download_re2_dataset,
    download_re3_dataset,
)

download_re1_dataset()  # 375 cases, metrics only
```

DRAFT

```
download_re2_dataset()  # 270 cases, metrics + logs + traces
download_re3_dataset()  # 90 cases, code-level faults
```

The datasets are available in the data directory after downloading. Details about the datasets are available in the Zenodo repository (DOI: 10.5281/zenodo.14590730).

Acknowledgements

We would like to express our sincere gratitude to the researchers and developers who created the baselines used in our library. Their work has been instrumental in making this project possible. We deeply appreciate the time, effort, and expertise that have gone into developing and maintaining these resources. This project would not have been feasible without their contributions.

This framework is built upon my previously published academic work (Pham et al., 2024a, 2024b, 2025) on RCA for microservice systems. As I am working toward general root cause analysis, this software paper positions the framework for the general RCA task without limiting itself to microservice systems. Future improvements to this framework will focus on the inclusion of RCA methods and datasets from different fields.

References

Pham, L., Ha, H., & Zhang, H. (2024a). BARO: Robust root cause analysis for microservices via multivariate bayesian online change point detection. *Proceedings of the ACM on Software Engineering*, *1*, 2214–2237. https://doi.org/10.1145/3660805

Pham, L., Ha, H., & Zhang, H. (2024b). Root cause analysis for microservices based on causal inference: How far are we? *Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering*, 706–715. https://doi.org/10.1145/3691620.3695065

Pham, L., Zhang, H., Ha, H., Salim, F., & Zhang, X. (2025). RCAEval: A benchmark for root cause analysis of microservice systems with telemetry data. *Companion Proceedings of the ACM Web Conference 2025*, 777–780. https://doi.org/10.1145/3701716.3715290

Xin, R., Chen, P., & Zhao, Z. (2023). CausalRCA: Causal inference based precise fine-grained root cause localization for microservice applications. *Journal of Systems and Software*, *203*, 111724. https://doi.org/10.1016/j.jss.2023.111724

DRAFT